

Chapter #4 & 5 Summative Test

Name: _____

Date: _____

Knowledge	Application	Thinking	Communication
/22	/14	/14	/5

KNOWLEDGE

Multiple choice (8 marks)

1. The exact radian measure of 105° is

- a. $\frac{5\pi}{11}$ b. $\frac{7\pi}{12}$ c. $\frac{5\pi}{24}$ d. $\frac{13\pi}{12}$

2. The exact degrees measure of $\frac{13\pi}{24}$ is

- a. 22.5° b. 75° c. 97.5° d. 142.5°

3. The exact value of $\cos \frac{5\pi}{3}$ is

- a. $\frac{\sqrt{3}}{2}$ b. $\frac{1}{2}$ c. $-\frac{\sqrt{3}}{2}$ d. $-\frac{2}{\sqrt{3}}$

4. Given that $\cot x = -1$ and the angle x lies in the second quadrant, what is $\sec x$?

- a. $\frac{1}{\sqrt{2}}$ b. $-\frac{1}{\sqrt{2}}$ c. $\sqrt{2}$ d. $-\sqrt{2}$

5. Determine an exact value for the expression $\cos \frac{2\pi}{3} \cos \frac{5\pi}{6} + \sin \frac{2\pi}{3} \sin \frac{5\pi}{6}$

- a. $\frac{1}{2}$ b. $\frac{\sqrt{3}}{2}$ c. $\frac{\sqrt{2}}{2}$ d. $\frac{\sqrt{2}}{3}$

6. What is the maximum value of the function $y = \cos x - 2$

- a. -1 b. -2 c. 0 d. 1

7. If a circle has a diameter of 10 and the central angle measure 60° , the arc length is

- a. $\frac{7\pi}{6}$ b. $\frac{5\pi}{6}$ c. $\frac{7\pi}{3}$ d. $\frac{5\pi}{3}$

8. The exact value of (-1) is

a. $\frac{7\pi}{4}$

b. $\frac{\pi}{4}$

c. $\frac{5\pi}{4}$

d. $-\frac{\pi}{4}$

9. Prove each of the following identities (6 marks)

a. $1 - \frac{2}{x+1} = \frac{x-1}{x+1}$

b. $\cos^2 x = \frac{\csc x \cos x}{\tan x + \cot x}$

10. For the function of $y = -2 \sin(\pi + 2x) + \frac{1}{2}$, determine the (6 marks)

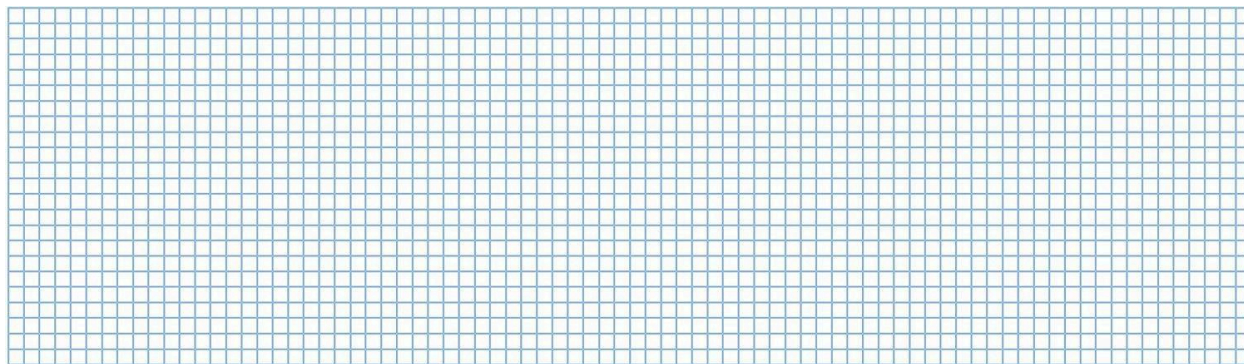
a. Amplitude: _____

b. The phase shift: _____

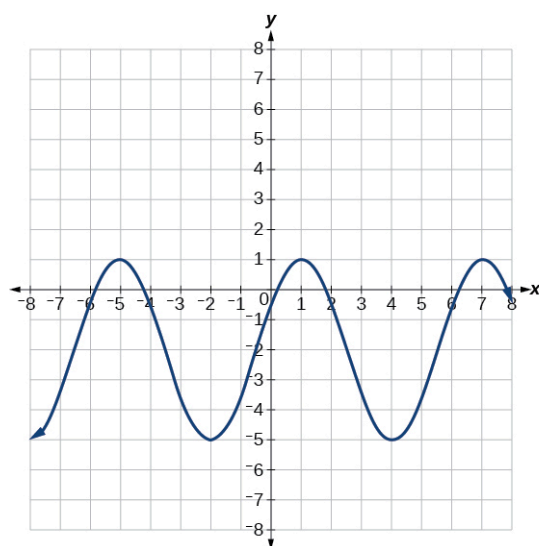
c. The period: _____

d. Vertical translation: _____

e. Sketch the function:

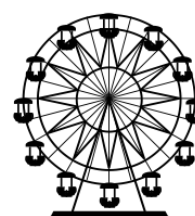


11. Determine one sine and one cosine function for the following function (2 marks)



APPLICATION

12. A Ferris wheel with radius of 12 meters is traveling at the speed of 2 minutes per rotation. The center of the wheel is 13 meters from the ground. (8 marks)



- Determine the equation of the height of a passenger above the ground, $h(t)$ in meters, and t in seconds, assuming that the passenger gets on at the lowest point at $t=0$.
- Determine the height of the passenger 30 seconds after boarding

- c. Determine the time when the passenger is 5 meters above the ground after 1 full rotation since the beginning of the ride

13. Given $\sin x = \frac{4}{5}$, $\left(\frac{\pi}{2} < x < \frac{3\pi}{2}\right)$ and $\cos y = \frac{5}{13}$, $(0 < y < \pi)$, find the exact values of each of the following (6 marks)

a. $\sin(x + y)$

b. $\cos 2x$

THINKING

14. Solve (6 marks)

a. $(\sin x - \cos x)^2 = \frac{1}{2}, \quad -\pi \leq x \leq \pi$

b. $\frac{1}{1+x} = -\cos x$

15. A cosine function has a maximum value of 3, a minimum value of -1, a phase shift of 1 radian to the right and a period of 4. Determine an equation for the function. (2 marks)

16. Determine all solutions for the following (6 marks)

a. $x - \cos x - 2 = 0$

b. $2x - 1 = 0$

COMMUNICATION

17. Explain how you would transform the function $y = \sin \sin x$ into $y = -\sin(2x - 3) - 5$. State *all* steps. Discuss verbally with the teacher. (3 marks)

18. Explain graphically why $\sin x = 0.75$ has 2 solutions for $0 \leq x \leq 2\pi$ (2 marks)

FORMULA SHEET

Co-Function Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\csc\left(\frac{\pi}{2} - x\right) = \sec x$$

$$\sec\left(\frac{\pi}{2} - x\right) = \csc x$$

$$\cot\left(\frac{\pi}{2} - x\right) = \tan x$$

Double Angle Formulas

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\cos(2x) = 2\cos^2 x - 1$$

$$\cos(2x) = 1 - 2\sin^2 x$$

Compound Angle Formulas

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$